

Proteins are macromolecules essential to all living organisms. Protein language models (PLMs) trained on protein sequences are used in bioinformatics to predict their properties or to generate novel proteins. However, the inner workings of these large neural networks are not well understood. In this thesis, we build on the InterPLM project, which uses sparse autoencoders to interpret the features of a masked PLM. We extend this approach to a generative PLM and compare the interpretability of its features with those of the masked model. The work introduces new feature analysis methods, demonstrates their practicality, and extends the interactive web application for exploring features by concept or protein of interest. Finally, we use these features to steer the model’s behavior during generation. This is achieved by clamping activations of selected features so as to induce the desired properties in the generated protein sequences. Analysis of the results shows that this kind of causal intervention is possible for a generative protein language model, as we demonstrate by generating disordered proteins and transit peptides.